A Two-Stage DDSP Acoustic Guitar Synthesizer Combining Differentiable Karplus–Strong and Modal Body Resonance

Zachary Burkett
Universitat Pompeu Fabra
zacharyjordan.burkett01@estudiant.upf.edu

Audio demos and code: https://github.com/Bottle-Water/DDSPAcousticGuitar/tree/main

Abstract

Abstract— Differentiable Digital Signal Processing (DDSP) enables parametric synthesizers to be trained from audio while retaining lightweight, interpretable controls at inference time [1]. In this work, we present a two-stage acoustic guitar synthesizer that combines a differentiable Karplus–Strong (KS) string model with a learned modal body resonator. For string synthesis, we build on the differentiable KS approach of Tablas de Paula et al., which uses Lagrange interpolation to make delay length (and therefore pitch) continuous and differentiable [3, 4]. To model the guitar body, we adopt a modal architecture inspired by Rau & Smith, using a parallel bank of second-order IIR resonators to capture invariant body resonances [2]. The system is trained in two stages: first fitting note-dependent string dynamics, then fitting shared body-mode gains with the string model frozen. This staged optimization enforces separation of string and body contributions and addresses the body-modeling limitation noted in prior differentiable KS work [3]. Preliminary results suggest that modal body filtering improves realism and low-note quality versus string-only synthesis.

1. Background

Differentiable Digital Signal Processing is a technique established by Engel et. al [1], where, by creating continuous parameter representations, DSP synthesis systems can be trained to accurately recreate audio samples, rather than directly generating audio. This has the significant advantage of, once trained, having a lightweight synthesizer that can be played in real time, and adjusted with meaningful parameters.

For a workshop, Pablo Tablas De Paula extended this paradigm to the Karplus-Strong algorithm, proposing a fully differentiable time-domain implementation [3, 4]. The major contribution is using Lagrange interpolation for fractional delay lengths, enabling pitch to be a continuous, differentiable parameter. They also include a DDSP IIR filter helper in the implementation. Notably, they identify the absence of body modelling as a key limitation, which particularly helps on the synthesis of lower notes.

Rau and Smith used physical modelling techniques and a modal filter to synthesize a resonator guitar, fitting M=24 modes as a parallel bank of second-order IIR filters [2]. Their results motivate a modal architecture for body modelling, where each resonance is characterised by a center frequency, damping ratio, and gain, a compact and physically grounded representation of body acoustics.

2. Motivation

This work directly addresses the gap identified by Pablo Tablas De Paula by pairing their differentiable KS string model with a learned modal body resonator, trained in two stages to ensure clean separation of string and body contributions. By developing a system to separate and learn body resonances, the research works towards developing a comprehensive DDSP acoustic guitar synthesizer. By having a DDSP acoustic guitar synthesizer, it opens the possibility of increasing the accessibility, portability, and ease of use of realistic acoustic guitar timbres. Furthermore, users would be able to create their own realistic acoustic guitar model, which they could then play as a VST, allowing them to have access to their guitar sound while composing, even if they do not physically have their guitar with them.

The work also takes steps towards architectural design within a DDSP system, establishing a new DDSP-framed methodology to address the task of separating resonant body harmonic contribution from sound sources, which may be applicable to further DDSP modelling applications, such as DDSP harmoniums, violins, and other resonant body instruments.

3. Method

3.1 Overview

The DDSP Acoustic Guitar synthesizer is a 2 stage synthesizer, consisting of a string synthesis through Karplus-Strong, and body resonance through a modal resonator bank. The system is trained in two distinct phases to ensure separation of harmonic contribution from the string and body systems.

3.2 Excitation: Karplus-Strong String Model

For the synthesis of the string sound itself, Karplus-Strong was chosen for its natural fit as a physical model of a vibrating string: a delay line whose length sets the fundamental frequency, with a feedback filter that replicates the energy loss and timbral damping of a real string over time. These are the parameters that the pitch conditioned MLP learns:

Gain is a scalar applied each time around the feedback loop, controlling how much energy is preserved per cy-

cle, directly setting the overall decay rate. Lower gain = faster decay. The mix scalar controls the blend between the current sample and the lowpass-filtered version of it in the feedback filter, so it determines how aggressively high frequencies are damped relative to lows. It can be understood as the coefficient of the one-pole filter mix=0 is no filtering (bright), mix=1 is maximum filtering (dark/muted).

A differentiable Karplus-Strong library has been created in prior work, using Lagrange interpolation to make the discrete delay line differentiable [3]. This work was used here, but careful engineering was required to sanely initialize and map the parameters of the unstable Karplus-Strong system to ensure training was smooth [5]. There is also a limitation of the library where a first-order low pass filter is used, severely limiting the accuracy of timbral reconstruction of lower pitches.

3.3 Intermediate Low Pass Filter

In between string and body there is a pitch conditioned low pass filter that is also learned within stage 1. This is currently used as an implementation to control the highs of the Karplus-Strong model without having to modify the existing library to utilize second-order filters in the delay line.

3.4 Pitch-Dependent Decay

On an acoustic guitar, higher fretted notes are created from further constraining the string length, or increasing string tension. Both of these approaches affect the decay of the note, as further constrained, higher pitched strings will have higher damping, and thus, decay faster. In order to accurately represent the pitch dependent decay of an acoustic guitar, a target f0 conditioned MLP is used, which controls both the mix and gain, as mentioned earlier.

3.5 Body Resonance: Modal Filter Bank

The original approach of learning a convolutional impulse response, while possible, was too unstable and computationally expensive. Instead, following Rau & Smith (2019) [2], the body is modeled as a parallel bank of 24 second-order IIR resonators, consistent with the mode count found in their physical admittance measurements. Center frequencies and band count were determined empirically, settling on 24 log-spaced bands between 80–8000 Hz.

3.6 Recording Protocol for Invariance

In order to separate the variant string harmonic content from the invariant body resonance, it was required to create a dataset of multiple notes played on the same guitar. Using a laptop microphone in an acoustically damped room, a 12 note dataset was created by recording each open string and 12th fret note on the same guitar. Using this dataset, the model is given the human annotated Hz of the target note, as well as the wav file (represented in both envelope and multi-resolution STFT for loss).

3.7 Two-Stage Training

As mentioned earlier, two stage training separates the harmonic content of the string and resonator systems. In stage one, the Karplus-Strong synth is unfrozen and learned without resonators. The loss is a combined log-RMS envelope matching loss and multiscale-STFT loss, to replicate both harmonic and dynamic content of the string.

Stage 2 freezes the parameters found in Stage 1, and operates solely on the gain levels of the 24 log-distributed static frequency resonator modal filter bank. The modal resonator parameters are found to fit invariantly across the entire dataset, finding the true resonances of the guitar body, instead of the string. By freezing the one stage while the other is learning, there is a forced separation of responsibility.

By requiring the model to learn the Karplus-Strong for each note independently, the harmonic content of the string can be learned per note.

4. Results

Audio demos and code are available here: `https://github.com/Bottle-Water/DDSPAcousticGuitar/tree/main`.

As seen in Figure 1, the system creates a good approximation of the target guitar sound. In the ablation, it can be seen that both the trained low pass filter and body resonances are essential components to controlling the highs of the synthesized sound. You can see the body resonators are doing more work in Figure 2, which validates Tablas de Paula's original claim that body modelling helps most on low notes. All attempted notes sounded consistent with a guitar-like sound. The model is also able to interpolate pitches in between the given training data.

There is some high frequency mismatch on lower notes, which is a direct limitation of the first order filter in the DiffKS library, and should be a target of future development. The model struggles with the complex attack of a guitar pluck, which could be worked towards with a layered enveloped noise DDSP module.

5. Conclusion / Reflection

5.1 Conclusion

Overall, the experiment was a success. There was significant difficulty in this project of mapping the fragile parameter ranges [5] of the Karplus-Strong system to learnable parameters, while retaining stability of the system. The contribution of the static resonant body training strategy was highly effective, and could be used in further applications. I learned how to create a DDSP system with DDSP building blocks, and I was happy with the success of my architectural and algorithmic decisions to solve the core problems of the pitch and resonance needing separation of harmonic contribution.

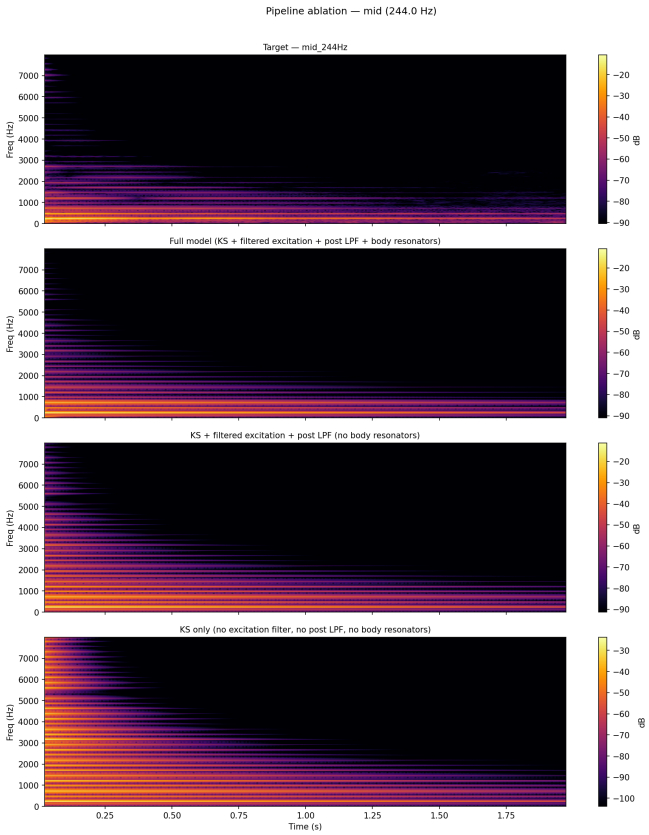

Figure 1: Ablation study at 244 Hz showing target, full model, KS + filters only, and KS-only outputs.

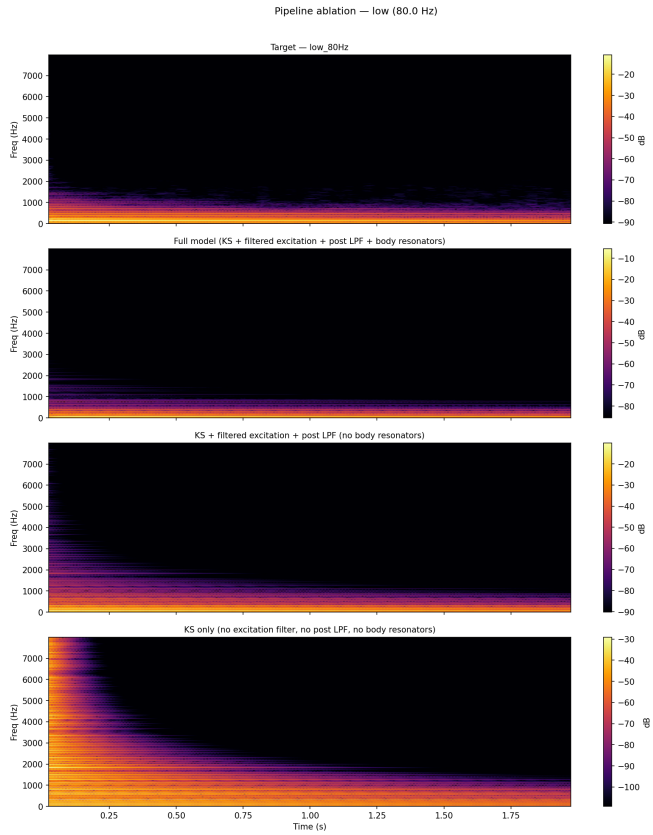

Figure 2: Ablation study at 80 Hz showing target, full model, KS + filters only, and KS-only outputs

5.2 Limitations

There is a limitation of the current approach, where each note is trained independently, with no relation to the actual string it was on. On a guitar, the same note will sound different depending on the string it is played on, due to the parameters (such as tension and damping) of the string being different across frets and strings. This suggests an extension to the model that is conditioned on a per-string and per-note basis, as opposed to the current per-note basis. This would allow for synthesis of the same pitch at multiple timbres, consistent with the behavior of a real acoustic guitar.

A natural extension to the work would be to allow the model to learn the frequencies themselves, as well as the gains of the modal filter bank. A second-order filter in the Karplus-Strong model itself would be more physically accurate and likely significantly improve model performance.

References

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